



ANDERSON JUNIOR COLLEGE
2015 JC 2 PRELIMINARY EXAMINATIONS

CHEMISTRY

9647/03

Higher 2

16 September 2015

Paper 3 Free Response Questions

2 hours

Candidates answer on separate paper.

Additional Materials: Data Booklet
 Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

Start each question on a fresh sheet of paper.

A Data Booklet is provided.

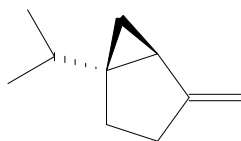
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of **12** printed pages.

1 This question is about the reactions of alkenes.

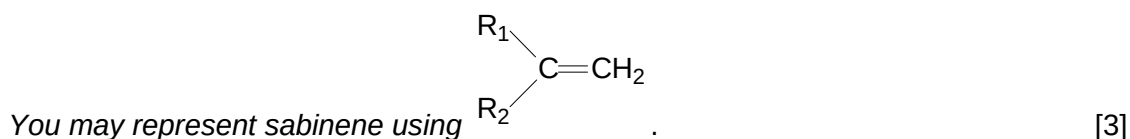
(a) Sabinene is an alkene that contributes to the flavor of nutmeg.



sabinene

Like other alkenes, it undergoes hydrohalogenation when treated with hydrogen halides.

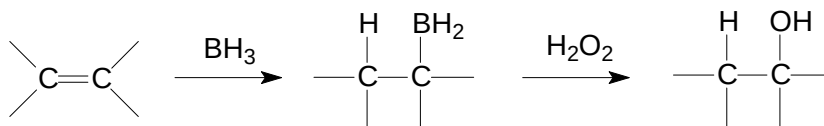
(i) Describe the mechanism of the reaction between sabinene and hydrogen chloride.



(ii) Hydrohalogenation of unsymmetrical alkenes results in a mixture of products. In such cases, the major product can be predicted using Markovnikov's rule.

With reference to your mechanism in (a)(i), account for the selectivity of the reaction. [2]

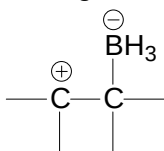
Hydration of alkenes via hydroboration favours formation of the anti-Markovnikov product. The hydroboration reaction involves 2 stages; first with limited borane, BH_3 , followed by treatment with alkaline hydrogen peroxide.



(b) (i) Explain why borane is considered an electron-deficient species. [1]

The initial reaction between the alkene and borane can be considered to occur in a similar fashion as the hydrohalogenation reaction in (a)(i).

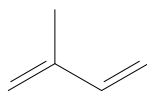
It is suggested that the mechanism goes through the formation of the intermediate below.



(ii) Suggest the mechanism showing the formation of the intermediate given above when sabinene reacts with borane. Indicate clearly the polarity of the B–H bond in borane by drawing δ^+ and δ^- on the appropriate atoms. [2]

(iii) By determining the change in the oxidation number of the reactive carbon, suggest the role of hydrogen peroxide in the reaction. [2]

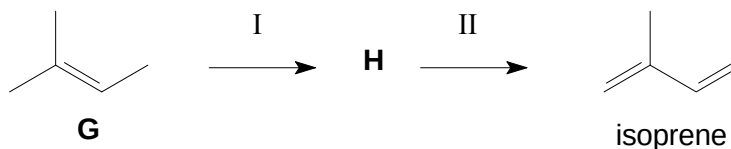
- (iv) Predict the structure of the major product formed when isoprene undergoes hydroboration.



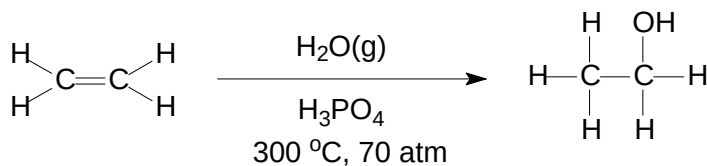
isoprene

[1]

- (c) Isoprene can be formed from compound **G** in 2 steps.



- (i) Identify compound **H**. [1]
- (ii) State the reagents and conditions for steps I and II. [2]
- (iii) Describe a chemical test to distinguish **G** and isoprene. [2]
- (d) Industrial preparation of ethanol involves the hydration of ethene with phosphoric acid catalyst.



- (i) Write an equation for the formation of phosphoric acid from phosphorous(V) oxide, P_4O_{10} , and water. [1]
- (ii) Describe the reaction, if any, that occur when separate samples of aluminum oxide and magnesium oxide are added to water. Write equations where appropriate and suggest the pH of any aqueous solution formed. [3]

[Total: 20]

- 2 It was first reported in 1853 that hydrogen peroxide can be manufactured by electrolysis of 10 mol dm^{-3} sulfuric acid, using a high current density and a shiny platinum anode.

(a) Sulfuric acid is a strong *Brønsted–Lowry acid* and it ionises in two stages.

- (i) Explain the term in *italics*. [1]
- (ii) Write equations to show the equilibria that exist in these **two** stages of ionisation. [1]
- (iii) Sulfuric acid is a strong acid and it may be assumed that the first stage of the ionisation is complete.

For a solution containing 10 mol dm^{-3} sulfuric acid, state the concentrations of hydrogen ion and hydrogen sulfate ion, HSO_4^- , due to this ionisation. [1]

- (iv) Write an expression for the acid dissociation constant, K_a , for the second stage (that is, the stage in which the hydrogen sulfate ion behaves as an acid). The value of this K_a is $1.0 \times 10^{-2} \text{ mol dm}^{-3}$. [1]
- (v) Calculate the total hydrogen ion concentration of 10 mol dm^{-3} sulfuric acid from both stages in the ionisation. [2]
- (vi) Hence, calculate the concentrations of other ions present in this solution. [1]
- (vii) Hydrogen sulfate ion can behave both as an acid and a base in aqueous solution. Using the information given, explain whether it is a stronger acid or base. [2]

It was later found that peroxodisulfuric acid, $\text{H}_2\text{S}_2\text{O}_8$, is formed initially during the electrochemical process before undergoing hydrolysis to form hydrogen peroxide and sulfuric acid.

- (b) (i) Suggest equations to show the likely reactions that occur at the two electrodes in the cell. [2]
- (ii) Write an equation for the hydrolysis of peroxodisulfuric acid to give hydrogen peroxide and sulfuric acid. [1]
- (iii) The mechanism of this hydrolysis is thought to proceed via two steps.
 - ☺ There is a partial hydrolysis of $\text{H}_2\text{S}_2\text{O}_8$ by H_2O molecule to form peroxomonosulfuric acid, H_2SO_5 .
 - ☺ This is followed by the attack of a second H_2O molecule on H_2SO_5 .

Use this information given to write **two** equations which represent the likely mechanism for the hydrolysis of peroxodisulfuric acid to hydrogen peroxide. [2]

- (iv) Draw the likely displayed formula of H_2SO_5 and on it show suggested values for all bond angles. [3]
- (v) The oxidation number of sulfur in $\text{H}_2\text{S}_2\text{O}_8$ is the same as in H_2SO_5 . Suggest the oxidation number of S in both compounds. [1]

- (c) The decomposition of hydrogen peroxide is very slow and can be catalysed by a homogeneous catalyst.

By considering relevant $E^\ominus$ values from the Data Booklet, show by means of equations, how $\text{Mn}^{2+}(\text{aq})$ ions can act as a homogeneous catalyst in this reaction. [2]

[Total: 20]

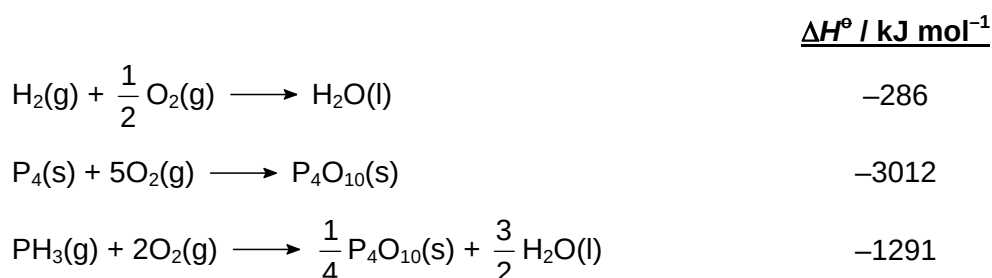
- 3 (a) Free volume, V , refers to the volume of space between gas molecules. For an ideal gas, free volume is essentially the same as the volume of the container. This can be calculated using the ideal gas equation.

$$pV = nRT$$

- (i) Under what conditions of temperature and pressure would you expect the behaviour of a real gas to be most like that of an ideal gas? [1]
- (ii) The pressure of a gaseous sample of 68 g of phosphine, PH_3 , in a container of volume 400 cm^3 is measured to be $3.36 \times 10^7 \text{ Pa}$ at 800 K.
- Using the ideal gas equation, calculate the free volume of this gaseous sample. Express your answer in cm^3 . [2]
- (iii) Explain why the volume you have calculated in (a)(ii) differs from that of the volume of the container. [2]

- (b) Phosphine is a colourless, poisonous gas that is commonly used as a fumigant.

- (i) Write an equation to represent the enthalpy change of formation of phosphine, PH_3 . [1]
- (ii) Using the data given below, calculate the enthalpy change of formation of phosphine.



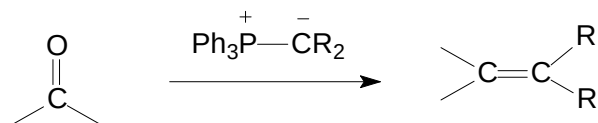
[2]

- (iii) Some bond energy values are given in the table below.

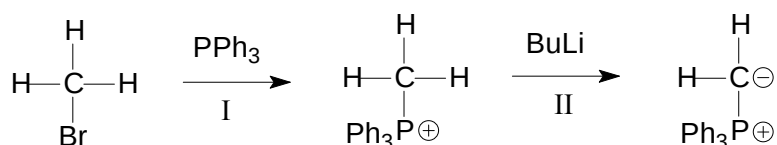
bond	bond energy / kJ mol^{-1}	bond	bond energy / kJ mol^{-1}
H–H	436	Cl–Cl	244
P–P	208	H–Cl	431
P–H	322	P–Cl	330

The P–H bond energy is the average of the H–H and P–P values. Explain why the H–Cl bond energy is **not** the average of the H–H and Cl–Cl values. [1]

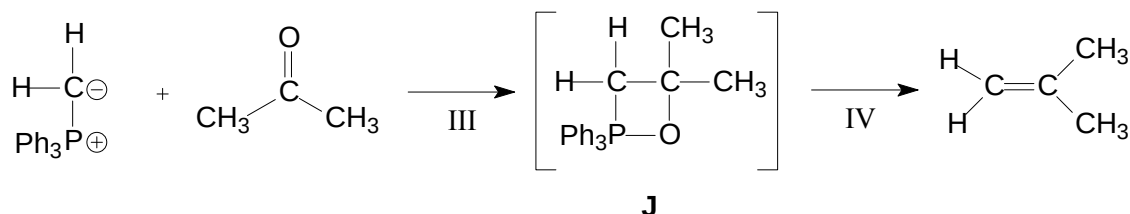
- (c) In 1954, Georg Wittig discovered that alkenes can be prepared from aldehydes and ketones using a phosphorus ylide $\text{Ph}_3\text{P}^+-\text{CR}_2^-$, where Ph represents a phenyl group.



An example of how 2-methylpropene is synthesised using the Wittig reaction is shown below. The phosphorus ylide is first synthesised from the reaction of bromomethane with triphenylphosphine, PPh_3 .



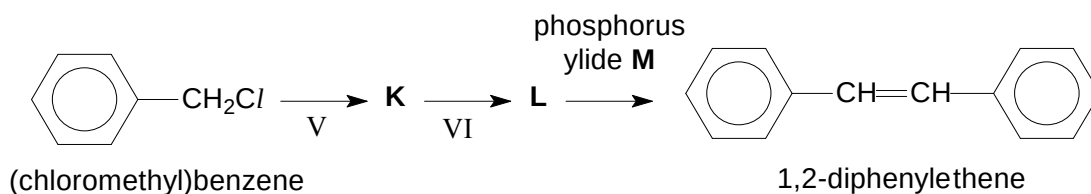
The phosphorus ylide is then reacted with propanone to form 2-methylpropene.



- (i) Suggest the type of reaction occurring in stage I. [1]
- (ii) State the role of butyl lithium, BuLi , in stage II. [1]
- (iii) Suggest the oxidation numbers of the two sp^2 hybridised carbons in 2-methylpropene, indicating clearly which carbon atom you are referring to. [2]
- (iv) In stage IV, besides 2-methylpropene, another by-product is also formed.

Suggest the mechanism for the formation of 2-methylpropene from intermediate **J**, showing curly arrows for the movement of electrons and the structure of the phosphorus-containing by-product that is formed. [2]

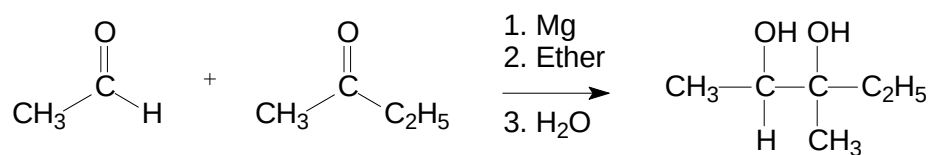
- (d) The Wittig reaction can be used as the last step in the following synthesis of 1,2-diphenylethene from (chloromethyl)benzene.



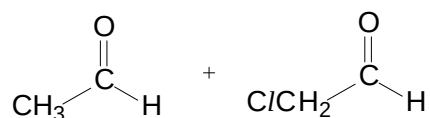
Suggest the reagents and conditions for steps V and VI, and the structures of the compounds **K**, **L** and phosphorus ylide **M**. [5]

[Total: 20]

- 4 (a) Carbonyl groups can be coupled together to form a diol.

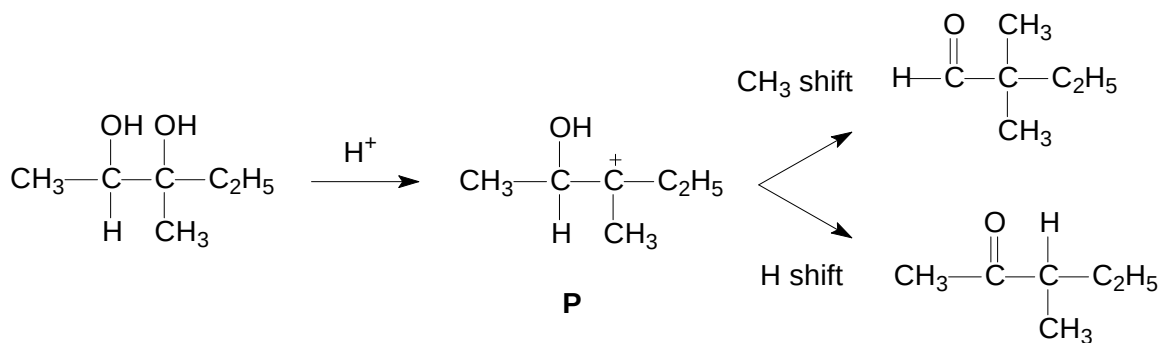


- (i) Draw the displayed formula of the diol that will be formed when the following compounds are reacted under the same set of conditions.

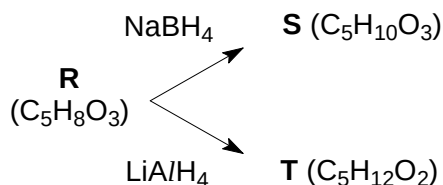


[2]

The Pinacol rearrangement involves the formation of carbonyl compounds from a diol via the following reaction scheme.



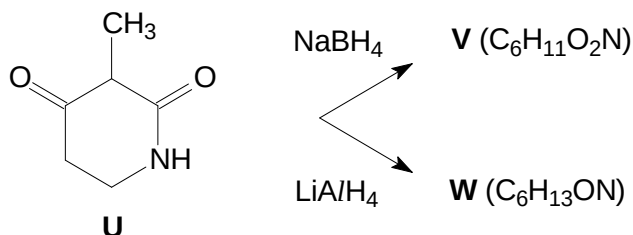
- (ii) Other than **P**, another carbocation, **Q**, can also be formed. Draw the structural formula of **Q** and suggest a reason why the yield of **Q** is smaller. [2]
- (iii) Draw the more stable carbonyl compound formed when the diol you have drawn in (a)(i) undergoes the Pinacol rearrangement, involving a H shift. [1]
- (b) LiAlH_4 and another complex hydride, NaBH_4 , are useful reducing agents in organic chemistry. LiAlH_4 is the more powerful of the two, as the following scheme shows.



All three compounds **R**, **S** and **T** are chiral and they give a yellow precipitate with alkaline aqueous iodine. Compounds **R** and **S** also react with $\text{Na}_2\text{CO}_3(\text{aq})$, but compound **T** does not.

- (i) Suggest structures for **R**, **S** and **T**. [3]

- (ii) The following scheme shows how compound **U**, which contains an amide group, behaves differently towards the two reducing agents.



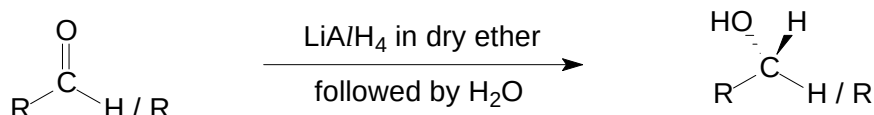
Both compounds **V** and **W** react with sodium metal. Only compound **W** reacts with HCl(aq) to give a salt.

Suggest structures for **V** and **W**. [2]

- (iii) State one reagent, other than sodium metal, that can distinguish between compounds **U** and **V** and describe what would be observed. [2]
- (iv) Suggest and explain how the basicity of ethylamine might compare to that of compound **U**. [2]
- (c) After the reduction is finished, excess LiAlH_4 is destroyed as follows.
- Water is added; hydrogen is evolved.
 - Dilute acid is added to dissolve the mixture of hydroxides.

Suggest equations for the reactions that occur during the two processes above. [2]

- (d) LiAlH_4 can be used to form alcohols from carbonyl compounds.



LiAlH_4 acts as a source of the hydride ion (H^-). The mechanism for the reduction of aldehydes and ketones is thought to involve two steps.

- ☺ The first step involves a nucleophilic attack of the hydride ion on the carbonyl carbon.
 - ☺ This is followed by an acid–base reaction to give the alcohol product.
- (i) Using the information given, describe the mechanism for the reduction of butanone by LiAlH_4 . In your answer, show any relevant charges, lone pairs of electrons and movement of electrons. [2]
- (ii) With reference to the mechanism you have described in (d)(i), explain why the alcohol formed does not rotate plane-polarised light. [1]
- (iii) Suggest why LiAlH_4 does not reduce alkenes to alkanes. [1]

[Total: 20]

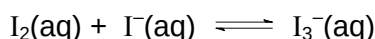
- 5 (a) The solubility of iodine in two immiscible phases at equilibrium can be measured by the *partition coefficient*, D where

$$D = \frac{[I_2]_{\text{solvent1}}}{[I_2]_{\text{solvent2}}}$$

A student carried out an experiment to determine the partition coefficient of iodine between tetrachloromethane and water. It was found that an aqueous solution containing $0.00120 \text{ mol dm}^{-3}$ of iodine was in equilibrium with a solution in tetrachloromethane containing $0.102 \text{ mol dm}^{-3}$ of iodine at room temperature.

- (i) Using the data given above, calculate the partition coefficient, D , of iodine between tetrachloromethane and water. [1]

Molecular iodine is not very soluble in water but it dissolves readily in aqueous solutions containing iodide ions. This is due to the formation of the complex triiodide ions, I_3^- , which exist in equilibrium with the iodine molecules, I_2 and iodide ions, I^- .



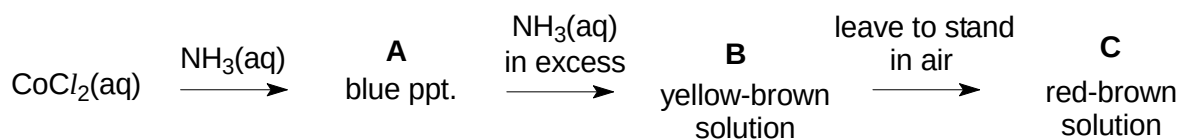
- (ii) The concentration of I_2 present in the equilibrium mixture decreases with increasing concentration of I^- . Explain why this is so. [1]

An experiment was carried out to investigate the equilibria represented above, where a solution of I_2 in $0.0232 \text{ mol dm}^{-3}$ of aqueous KI was shaken with tetrachloromethane until equilibrium was reached at room temperature.

The aqueous layer was found to contain I_2 and I_3^- at a total concentration of $0.0100 \text{ mol dm}^{-3}$ while the tetrachloromethane layer contained $0.0850 \text{ mol dm}^{-3}$ of I_2 .

- (iii) Use your value of partition coefficient, D , to calculate the concentration of molecular I_2 in the aqueous layer. [1]
- (iv) Hence, calculate the concentration of $I_3^-(aq)$ and $I^-(aq)$ in the aqueous layer. [2]
- (v) Calculate the value of K_c for this reaction. Include units in your answer. [2]

- (b) Aqueous cobalt(II) chloride, CoCl_2 is a pink solution which gives the following reactions.



- (i) What is the formula of the blue precipitate **A**? [1]
- (ii) By using the data given below and appropriate value from the *Data Booklet*, explain the observation when solution **B** is left to stand in air. [2]



- (iii) When concentrated hydrochloric acid is added to aqueous cobalt(II) chloride, the solution changes from pink to blue. When water is added to this blue solution, it becomes pink.

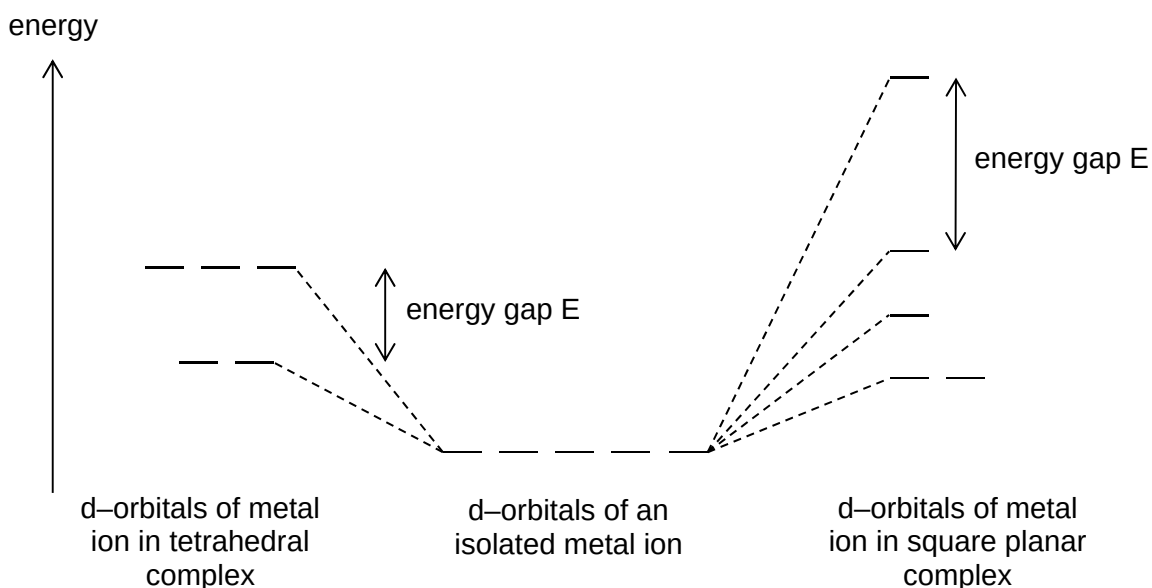
Suggest by means of an ionic equation, an explanation for these observations. [2]

- (c) Two different cobalt(III) complexes with the formula $\text{Co}(\text{NH}_3)_6\text{Cl}_3$ can be prepared under different conditions. Each complex contains a cobalt cation with a coordination number of 6. The complexes may be distinguished by their colours and by the amount of PbCl_2 precipitated when excess aqueous $\text{Pb}(\text{NO}_3)_2$ is added to aqueous solutions containing each of the complexes.

complex	colour	no. of moles of PbCl_2 precipitated per mole of complex	does the complex have a dipole moment?
E	purple	1.0	yes
F	green	0.5	no

- (i) Explain why an aqueous solution of **F** is green in colour. [3]
- (ii) Using the information given in the table, draw the structure of the cation in **F**, showing the arrangement of the ligands. [2]

- (d) Complexes with a coordination number of 4 can take on either the tetrahedral or the square planar structure. The following diagram shows how the d-orbitals of the metal ion are split in such complexes.



The arrangement of electrons in the d-orbitals depends on the spin states of complexes.

In a '*high spin*' state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

In a '*low spin*' state, the lower energy d-orbitals are filled first, by pairing up if necessary, before the higher energy d-orbitals are used.

- (i) Electrons usually prefer to occupy orbitals singly because of repulsion that would occur between two electrons occupying the same orbital.

In light of this, suggest why a complex might still adopt a *low spin* state. [1]

- (ii) Many transition elements and their compounds are paramagnetic, which means that they are attracted to a magnetic field. This property is due to the presence of unpaired electrons in the d orbitals.

The table below shows the relative paramagnetism of two complexes with different geometry, which contain Ni^{2+} ion with a coordination number of 4.

formula of complex	relative paramagnetism
$[\text{NiCl}_4]^{2-}$	2
$[\text{Ni}(\text{CN})_4]^{2-}$	0

Use appropriate diagrams from above, together with the information given above concerning the relative paramagnetism to show the electronic distribution of Ni^{2+} ion in $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$. [2]

[Total: 20]